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T.Y. Minor Project Presentation

AI-Driven Vulnerability Index Mapping for Districts in Maharashtra to El Niño Droughts Using Socioeconomic and Climatic PCA

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Introduction

The Global Phenomenon: El Niño is the unusual warming of the eastern Pacific Ocean that disrupts global atmospheric circulation.

The Impact on India: It weakens the monsoon, causing rainfall deficits and prolonged dry spells.

The Cascading Threat: Rainfall failure leads to crop failure, water scarcity and economic losses for farming communities.

True Vulnerability: It depends on both the climatic hazard and the socio-economic ability of people to cope.

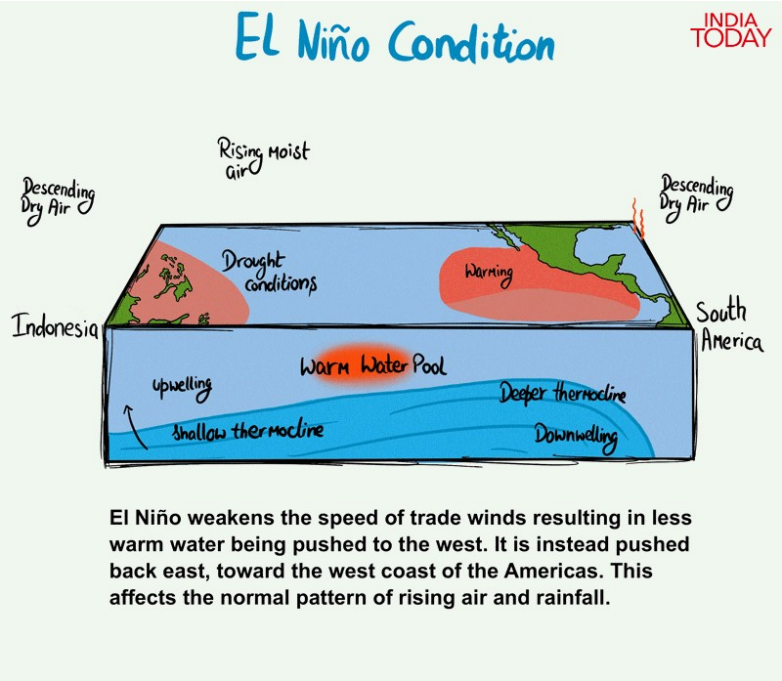


Fig. 1: Illustration of the El Niño climatic mechanism (Source:India Today)

Problem Statement & Proposed Solution

Problem Statement

- Existing drought indices often rely on subjective (AHP) weights and ignore socio-economic conditions at district level.
- The district-specific impact of El Niño on Maharashtra's monsoon is rarely quantified.
- An objective, data-driven and validated district vulnerability index is needed.

Proposed Solution

- Six direction-aligned indicators of El Niño exposure, agricultural sensitivity and adaptive capacity for 34 districts.
- PCA-derived weights combine them into a 0–100 index; K-Means (K = 3) gives Low, Medium and High groups.
- Validation against drought declarations (2018, 2023, 2026 holdout), with maps and an interactive dashboard.

Literature Survey

Sr. No.	Name of the Publication / Journal	Title of the Research Paper	Objective / Problem Discussed	Methodology	Dataset / Tools Used	Key Findings	Research Gap
1	Hydrology Research (2026)	Sustainable drought resilience strategies in India	Review national drought frameworks	Literature Review	Policy & Climate data	60% districts face water stress by 2030	Conceptual only; lacks local quantitative metrics
2	Environ Dev Sustain (2025)	Household vulnerability index to drought in Tamil Nadu	Assess rural household vulnerability	PCA	Surveys & Climate data	Ramanathapuram most vulnerable (0.86)	Limited to TN; no Maharashtra district mapping
3	Global NEST Journal (2024)	Maharashtra River Basin flood-drought nexus	Analyze river basin water scarcity	Qualitative Case Study	Rainfall & Urban metrics	Found distinct regional scarcity patterns	Lacks quantitative indexing & ML clustering
4	Water - MDPI (2023)	Drought vulnerability in NW Odisha via Fuzzy-AHP	Map basin-level drought risk	Fuzzy-AHP	GIS & Demographic data	Mapped 33% area as high drought risk	Subjective AHP weights; no unsupervised ML

Table 1: Literature Survey

Literature Survey

Sr. No.	Name of the Publication / Journal	Title of the Research Paper	Objective / Problem Discussed	Methodology	Dataset / Tools Used	Key Findings	Research Gap
5	Food Security (2023)	2023-24 El Niño impact on India food security	Evaluate El Niño agricultural risk	Economic Modeling	ONI & Crop yield data	>70% districts suffer crop yield drops	Macro national scope; lacks district mapping
6	Stoch Environ Res Risk Assess (2023)	Ensemble ML-based drought index in Odisha	Predict village drought vulnerability	Ensemble ML (M5P)	53 Spatial indicators	High risk in 37.9% sampled regions	Complex model; no unsupervised clustering
7	Gondwana Research (2022)	National drought vulnerability assessment	Map multidimensional drought risk	AHP & ML Regression	Climate & Census data	Socio-economic drought threatens food security	Manual weighing; no unsupervised ML grouping
8	Arab J Geosci (2022)	Drought vulnerability in Vidarbha, Maharashtra	Assess regional drought intensity	SPI & SPEI Indices	Historic IMD rainfall	High drought severity in East Vidarbha	Index-based; stopped short of K-Means

Table 1: Literature Survey (Continued)

Literature Survey

Sr. No.	Name of the Publication / Journal	Title of the Research Paper	Objective / Problem Discussed	Methodology	Dataset / Tools Used	Key Findings	Research Gap
9	J King Saud Univ Sci (2022)	Drought vulnerability in Kangsabati Basin	Delineate basin hazard zones	GIS-based AHP	Remote Sensing & GIS	Mapped 21.1% area as high risk	Subjective AHP bias; lacks objective weighting
10	Geomatics Nat Haz Risk (2021)	Spatial drought assessment via Fuzzy-AHP	Map state-wide vulnerability	Fuzzy-AHP	Weather & Census data	33.94% area mapped as high risk	Expert-dependent weights; no ML clustering
11	Natural Hazards (2021)	Regional drought vulnerability and risk using PCA and GMM	Objectively weight drought vulnerability indicators	PCA, GMM, equal weighting	Chungcheong Province, South Korea	PCA gave larger weights to agricultural and industrial factors	Not El Niño-specific; outside India
12	PLOS One (2026)	Territorial composite vulnerability index, Cali	Guide public health action by territory	PCA weights from eigenvalue and absolute loading	7 socio-economic indicators, Cali, Colombia	3 components (67.9% variance); KMO 0.60	Urban health focus; no climate hazard

Table 1: Literature Survey (Continued)

Research Gaps

- Existing review and national-scale studies are largely conceptual and do not provide district-level drought vulnerability mapping for Maharashtra.
- AHP and Fuzzy-AHP based indices depend on subjective expert judgement to assign the weights of indicators.
- Household-level and ensemble machine-learning indices focus on other states, use complex models and do not group districts through unsupervised learning.
- SPI and SPEI hazard indices and qualitative basin studies capture the drought hazard but do not integrate socio-economic and agricultural factors.
- Most existing studies do not measure El Niño-specific exposure and offer limited validation against observed drought declarations.
- This project addresses these gaps through an objective, PCA-weighted district index for Maharashtra that is classified with K-Means and validated against drought declarations.

Project Objectives

- Combine district-level climatic and socio-economic records into a unified dataset for Maharashtra.
- Apply PCA to objectively identify and weight the most critical vulnerability factors.
- Implement K-Means clustering to categorize districts into Low, Medium, and High-risk drought zones.
- Develop an interactive dashboard to map risk zones and enable proactive disaster management.

Study Area & Datasets

Dataset	Source	Variables Used	Period
Gridded daily rainfall (0.25°)	IMD	District June–September rainfall (area-weighted)	1950–2024
RONI / ONI	NOAA CPC	El Niño monsoon years (18), seasonal index values	1950–2024
Primary Census Abstract	Census of India	Agricultural workers, literacy (age 7+)	2011
Village Directory	Census of India	Irrigated and unirrigated area	2011
Agriculture Census	Govt. of Maharashtra	Holdings below 2 ha	2010-11
District boundaries	Survey of India	36 district polygons	Current
Drought GRs	Govt. of Maharashtra	Declared talukas (validation only)	2018, 2023, 2026

Table 2: Datasets Used

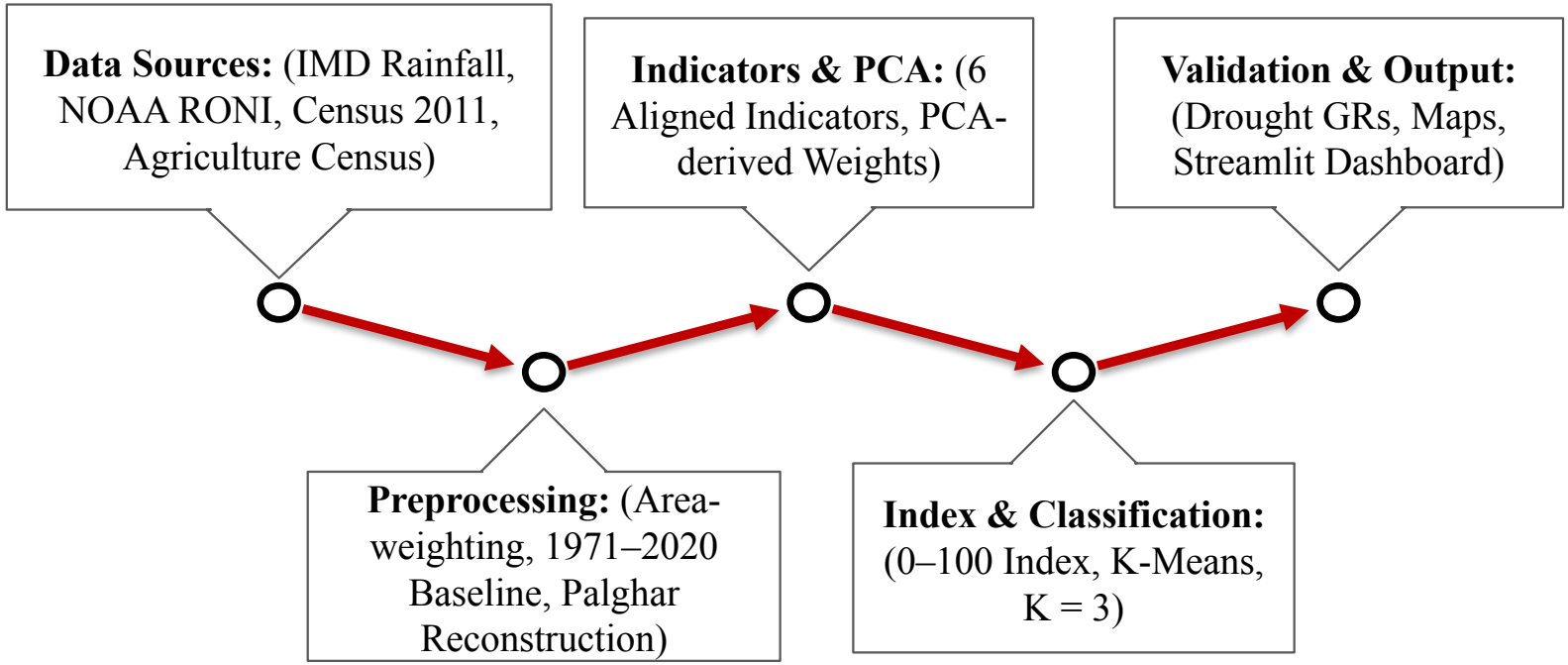
Selected Indicators

Study area: 36 districts of Maharashtra; 34 in the index (Mumbai City and Mumbai Suburban excluded, no agricultural data).
Selection: six indicators chosen from 30 candidate variables; all standardised (z-scores) before PCA.

Sr. No.	Indicator	Dimension	Direction Alignment	Source
1	El Niño rainfall sensitivity (% per °C RONI)	Exposure	× (−1): more negative slope = greater rainfall loss	IMD + NOAA RONI
2	El Niño deficient-monsoon frequency	Exposure	× (+1)	IMD + NOAA RONI
3	Agricultural-worker share of workers (%)	Sensitivity	× (+1)	Census 2011
4	Rainfed share of cropped land (%)	Sensitivity	× (+1)	Census 2011 (Village Directory)
5	Small-holding share (< 2 ha, %)	Sensitivity	× (+1)	Agriculture Census 2010-11
6	Literacy rate, age 7+ (%)	Adaptive capacity	× (−1): higher literacy = lower vulnerability	Census 2011

Table 3: Selected Indicators

Proposed Methodology



Architecture Diagram

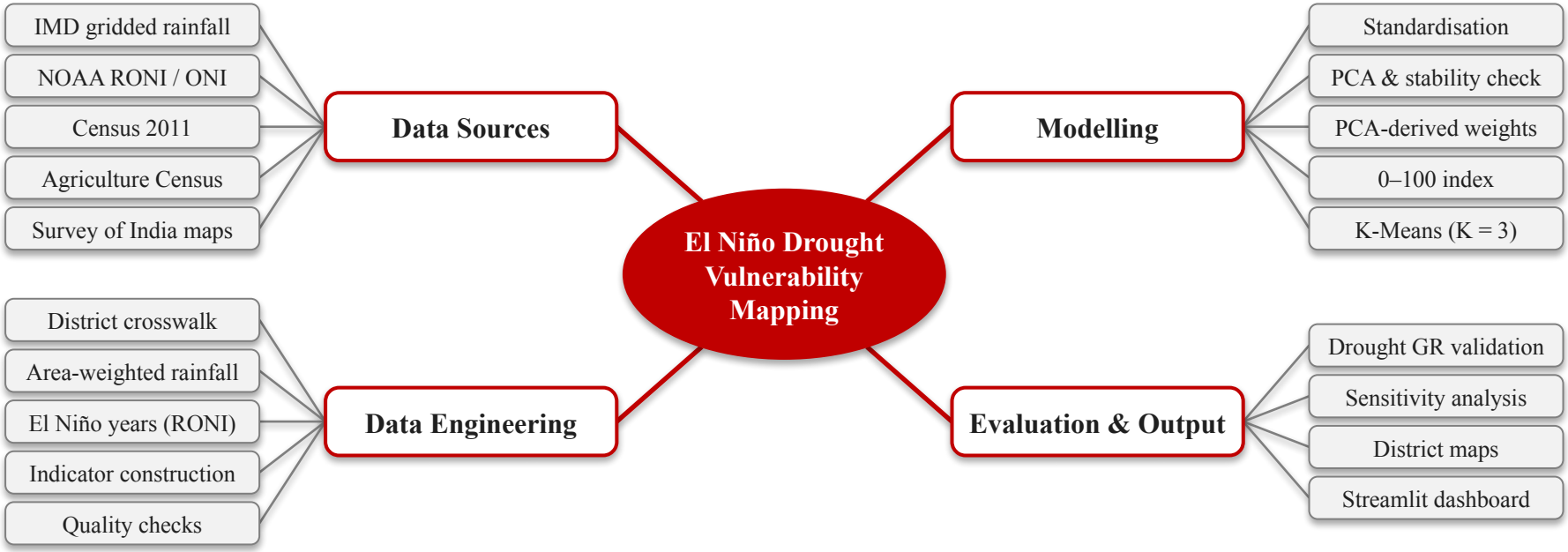


Fig. 2: Architecture of the Vulnerability Mapping Pipeline

Tools, Technologies and Resources

Hardware Requirements

- **Standard PC/Laptop:** Minimum 8 GB RAM with an Intel i5 / Ryzen 5 processor; about 2 GB storage for IMD gridded rainfall files.

Software & Tech Stack

- **Programming Language:** Python 3.12.
- **Data Processing & Modelling:** Pandas, NumPy, SciPy, Scikit-learn (StandardScaler, PCA, K-Means), xarray / netCDF4.
- **Geospatial & Visualisation:** GeoPandas, Shapely, Matplotlib; Streamlit for the interactive dashboard.
- **Quality & Version Control:** pytest (69 automated tests), Git and GitHub.

Datasets (Maharashtra)

- **Climatic:** IMD 0.25° gridded daily rainfall (1950–2024); NOAA CPC RONI and ONI.
- **Socio-economic & Agricultural:** Census 2011 (Primary Census Abstract, Village Directory); Agriculture Census 2010-11.
- **Spatial & Validation:** Survey of India district boundaries; Government of Maharashtra drought GRs (2018, 2023, 2026).

PCA Results & Weights

Component	Eigenvalue	Explained Variance (%)	Cumulative (%)
PC1	2.056	34.3	34.3
PC2	1.246	20.8	55.0
PC3	1.165	19.4	74.5
PC4	0.663	11.1	85.5
PC5	0.616	10.3	95.8
PC6	0.253	4.2	100.0

Table 4: PCA Eigenvalues and Explained Variance

Indicator	Dimension	Weight
El Niño rainfall sensitivity	Exposure	0.168
El Niño deficient frequency	Exposure	0.151
Agricultural-worker share	Sensitivity	0.183
Rainfed share of cropped land	Sensitivity	0.173
Small-holding share	Sensitivity	0.161
Literacy (aligned)	Adaptive capacity	0.164
Total		1.000

Table 5: PCA-Derived Indicator Weights

- First three components (eigenvalue > 1, 74.5% variance) are retained; each weight combines explained variance and absolute loadings, normalised to 1.
- KMO = 0.40 indicates weak common-factor structure, so the weights are exploratory and close to equal.

K-Means Classification

- K-Means (K = 3, random_state = 2026, n_init = 20) applied only to the 0–100 index; clusters ordered by centroid (lowest → Low).
- Silhouette score = 0.587; group boundaries at index values 29.8 and 64.1.

Group	Districts	Centroid	Index Range	Member Districts
High	10	82.4	65.9–100.0	Beed, Jalna, Latur, Dharashiv, Parbhani, Nanded, Nandurbar, Ahilyanagar, Solapur, Hingoli
Medium	20	45.9	32.6–62.7	Washim, Buldhana, Satara, Sangli, Dhule, Akola, Chh. Sambhajinagar, Palghar, Ratnagiri, Kolhapur, Jalgaon, Amravati, Yavatmal, Sindhudurg, Gadchiroli, Pune, Nashik, Raigad, Chandrapur, Wardha
Low	4	13.8	0.0–25.6	Bhandara, Gondia, Thane, Nagpur

Table 6: K-Means Vulnerability Groups

Results: Vulnerability Maps

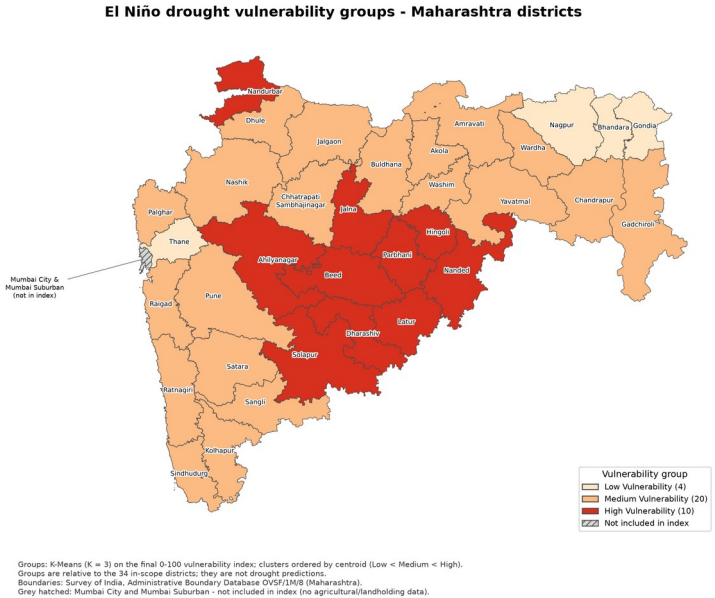


Fig. 3: Low / Medium / High Vulnerability Groups

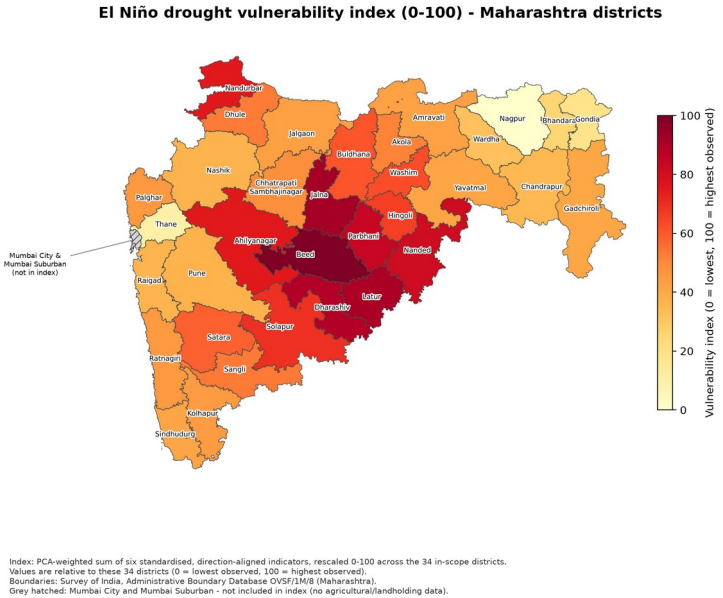


Fig. 4: Vulnerability Index (0–100)

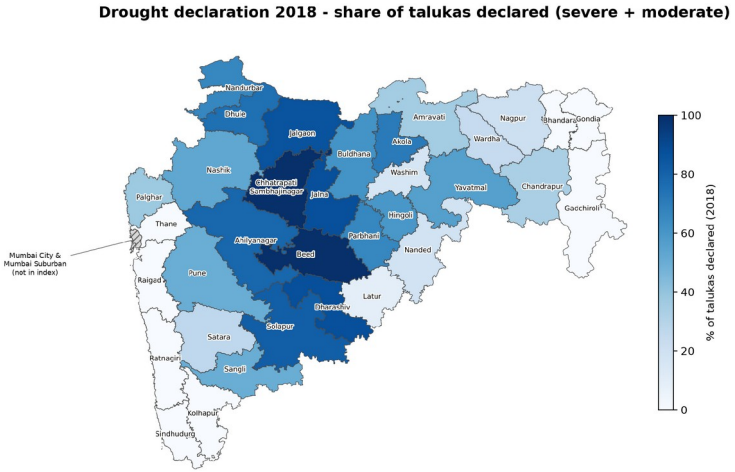
Validation against Drought Declarations

Year	Declaration (Govt. of Maharashtra GR)	Spearman ρ	p-value	Mean Declared Share (Low / Medium / High)
2018	151 talukas, final (severe + moderate)	0.585	0.0003	0.05 / 0.39 / 0.66
2023	40 talukas, final (severe + moderate)	0.450	0.008	0.00 / 0.10 / 0.20
2026 (holdout)	265 talukas, Trigger-1 stage	0.545	0.0009	0.39 / 0.70 / 0.99

- Declared share = declared talukas ÷ Census 2011 talukas per district (n = 34); declarations were never used to build the index.
- Higher index values were positively associated with larger declared shares in all three years, including the unseen 2026 holdout.
- Limitations: the 1,021 separately listed revenue circles of 2023 are not included; the 2026 Trigger-1 stage has a ceiling effect.

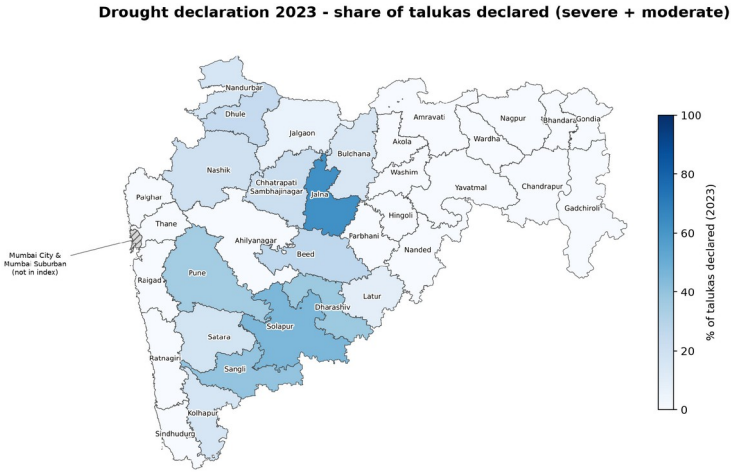
Table 7: Validation Results

Validation Maps (2018 and 2023)



Source: Government of Maharashtra GR 31-10-2018 (151 talukas). Government-declared share of each district's talukas (Census 2011 taluka count); not a direct measure of crop loss or rainfall.
Boundaries: Survey of India, Administrative Boundary Database OVSF/1M/B (Maharashtra).
Grey hatched: Mumbai City and Mumbai Suburban - not included in index (no agricultural/landholding data).

Fig. 5: Declared Taluka Share, 2018



Source: Government of Maharashtra GR 31-10-2023 (40 talukas). Government-declared share of each district's talukas (Census 2011 taluka count); not a direct measure of crop loss or rainfall.
Shows the main-order taluka declarations only; the 1,021 separately listed revenue-circle declarations (GR 10-11-2023) are not included.
Boundaries: Survey of India, Administrative Boundary Database OVSF/1M/B (Maharashtra).
Grey hatched: Mumbai City and Mumbai Suburban - not included in index (no agricultural/landholding data).

Fig. 6: Declared Taluka Share, 2023

Validation: 2026 Holdout

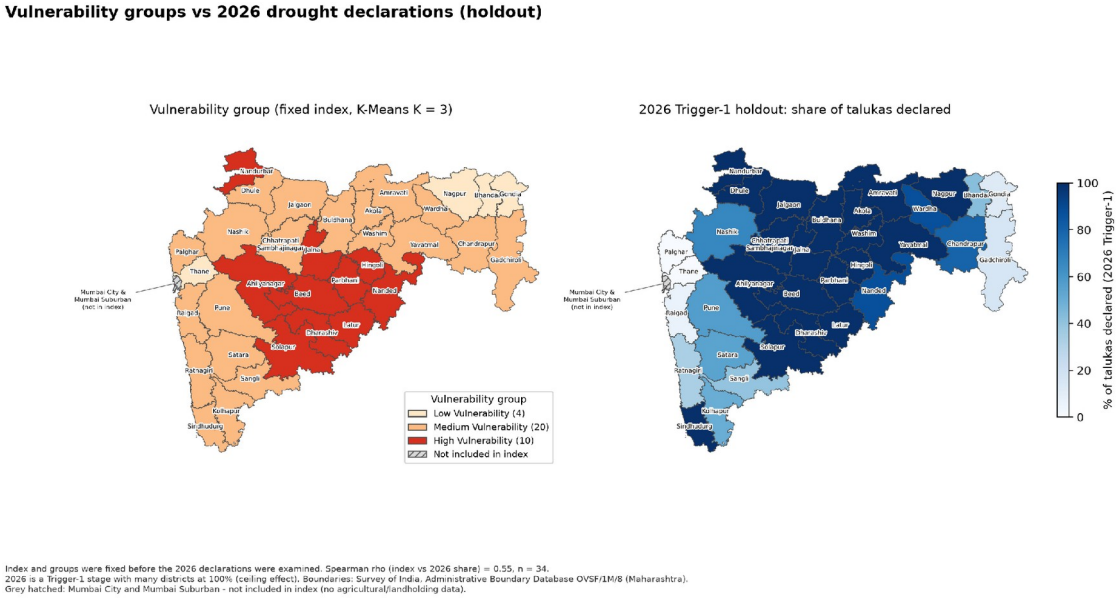


Fig. 7: Vulnerability Groups vs 2026 Trigger-1 Declarations (Spearman ρ = 0.545)

Robustness & Sensitivity Analysis

Comparison Version	Spearman ρ with Final Index	Largest Rank Change	Districts Changing Group
Equal weights (1/6 each)	0.993	5	2
PCA absolute-loading weights, all 6 components	0.995	3	1
Pune flagged rainfall years excluded	0.995	3	1
Hazard + social sub-indices (50/50)	0.991	6	1
ONI v6 instead of RONI	0.961	6	4
OECD/JRC factor weights	0.934	10	2

- K-Means silhouette for K = 2 / 3 / 4: 0.594 / 0.587 / 0.571; K-Means and a tertile split agree for 25 of 34 districts.
- No district moves between Low and High under any alternative; changes occur only near group boundaries.

Table 8: Robustness of the Index and Classification

Limitations

- Small sample of 34 districts: PCA and correlations carry statistical uncertainty, and parallel analysis retains only one component.
- Weak common-factor structure (KMO = 0.40): the PCA-derived weights are exploratory and close to equal.
- Socio-economic data from Census 2011 and Agriculture Census 2010-11 may not reflect current conditions.
- District averages mask taluka and village-level variation; suspect IMD rainfall values (e.g., Pune 2005) are retained as published.
- Drought declarations are administrative outcomes, so validation indicates association and not causation.

Future Scope

- Extend the analysis to taluka level and to other states or all of India.
- Update the indicators with newer census and agriculture census data as they are released.
- Incorporate additional climate drivers such as the Indian Ocean Dipole and La Niña, along with groundwater and heat stress.
- Validate against crop-yield losses and revenue-circle-level declarations, and update the index annually.

Conclusion

- A reproducible, data-driven El Niño drought vulnerability index was developed for 34 districts of Maharashtra using climatic, agricultural and socio-economic indicators.
- PCA provided objective indicator weights, and K-Means grouped the districts into Low (4), Medium (20) and High (10) vulnerability.
- High relative vulnerability is concentrated in Marathwada and adjoining dry districts, and is lowest in Nagpur, Thane, Gondia and Bhandara.
- The index is robust to alternative choices and positively associated with drought declarations, including the 2026 holdout; it is a relative, exploratory index, not a drought forecast.

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